

Education

Duke University

Durham, NC

B.S. Computer Science. Minors in Mathematics & Statistics

2023 - 2027

- **GPA:** 3.76
- **Coursework:** Foundations of Generative Models, Deep Learning, Natural Language Processing, Generative AI in Protein Design, Linear Algebra, Advanced Probability, Stochastic Processes, Data Structures & Algorithms, Databases
- **Dr. Bart Kamen Memorial Scholarship:** \$40,000 merit scholarship awarded to students with high research output
- **Volunteer Services:** Reviewer for NeurIPS 2025 Conference

Select Peer-Reviewed Conference & Journal Publications (Refer Google Scholar)

Token-Level Guided Discrete Diffusion for Membrane Protein Design

Shrey Goel, Perin Schray, Yinuo Zhang, Sophia Vincoff, Huong Kratochvil, Pranam Chatterjee.

Neural Information Processing Systems (NeurIPS) AI4Science Workshop. 2025. **[ORAL]**

FusOn-pLM: a fusion oncoprotein-specific language model via adjusted rate masking

Sophie Vincoff, **Shrey Goel**, Kseniia Kholina, Rishab Pulugurta, Pranay Vure, Pranam Chatterjee.

Nature Communications. 2025.

Minimal-Action Discrete Schrödinger Bridge Matching for Peptide Sequence Design

Shrey Goel, Pranam Chatterjee.

International Conference on Learning Representations (ICLR) DeLTA Workshop. 2026.

Rapid human oogonia-like cell specification via transcription factor-directed differentiation

Merrick Smela, Christian Kramme, Patrick Fortuna, Bennett Wolf, **Shrey Goel**, ... Pranam Chatterjee.

EMBO Reports. 2025.

Research Experience

Machine Learning Research

Cambridge, MA

Massachusetts Institute of Technology (Advisor: Dr. Connor Coley)

May 2026 – Present

- Developing and training a multimodal diffusion model combining SE(3) frame, all-atom, and discrete amino-acid diffusion.
- Designed RDKit graph search algorithms to assess quality of generated samples.

Machine Learning Research

Durham, NC

Duke University (Advisor: Dr. Pranam Chatterjee)

April 2023 – Present

Membrane Protein Diffusion Model – See Publication 1, Invited Talks 1-2

- Developed and trained MemDLM, a discrete diffusion model for membrane protein sequence generation using PyTorch and a HPC cluster.
- Generated sequences match performance of natural controls in wet-lab assays and reduced perplexity by 44% over state-of-the-art autoregressive models.
- Designed and implemented novel classifier-guided sampling algorithm combining attention scores and classifier gradients to selectively edit specific sequence tokens during inference.

- **First-author paper** published and selected for **oral presentation** at **NeurIPS 2025** AI4Science workshop.

Cancer Protein Representation Learning – [See Publication 2](#)

- Fine-tuned ESM-2 protein language model on a novel masked language modeling objective that leverages dynamic masking rates to engineer cancer protein-specific sequence embeddings.
- **Second-author manuscript** published in *Nature Communications*

Discrete Peptide Sequence Generation – [See Publication 3](#)

- Formulated and implemented MadSBM, a discrete sequence-generation framework based on Schrödinger Bridge Matching.
- Trained a Diffusion Transformer model in PyTorch and showed that MadSBM remained in high-probability regions relative to state-of-the-art diffusion and flow-matching models.
- **First-author manuscript** published at ICLR 2026 generative models workshop.

Bioinformatics Research

Duke University (Advisor: Dr. Pranam Chatterjee)

Durham, NC

April 2023 – Feb 2025

[See Publication 4](#)

- Reduced experimental analysis time from 4 months to 3 days by training language models for cell type classification.
- Created automated statistical analysis pipeline in R and Python to plot PCAs, volcano plots, and heat maps.
- Co-author manuscript published in *EMBO Reports*.

Professional Experience

Machine Learning Engineer

Latus Bio

Durham, NC

Sept 2025 - March 2026

- Trained sequence classifiers and multi-objective discrete flow matching models for AAV design.

Machine Learning Engineer Intern

Qualcomm

San Diego, CA

May 2025 - Aug 2025

- Diagnosed performance bottlenecks in Meta’s Llama LLM using PyTorch Profiler, identifying redundant computations over padded tokens in KV-cache system
- Engineered optimized self-attention tensor computations that bypass pad tokens, reducing inference latency by 14x in quantized Llama models deployed on edge-devices.

Invited Talks

Genentech (New York, New York) July, 2026
 NeurIPS AI4Science (San Diego, California) December, 2025

Community Service

- **Camp Kesem:** organized \$40,000 for summer camp for kids affected by their parent’s cancer.
- **Project Tadpole:** redesigned a ride-on toy car for a child with muscular dystrophy.

Technical Skills

Languages: Python, C++, Java, R, MIPS Assembly

ML Technologies: PyTorch, Hugging Face, Scikit-learn, PEFT, HPC, Pandas, NumPy